

P1: DROUGHT SEVERITY MAPPING USING VEGETATION CONDITION INDEX (VCI)

**Project Report Submitted in fulfillment of the Requirements
for the Award of the Internship of Summer Training
Program
Space Science Technology**

Enrolled Subject : Remote Sensing & GIS Using Python

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1. Title

Drought Severity Assessment Using Multi-Temporal Vegetation Condition Index (VCI) Derived from Satellite Data

2. Objective

The primary objective of this study is to assess the severity of drought across the Bundelkhand region by utilizing multi-temporal vegetation information obtained from satellite observations. The Vegetation Condition Index (VCI) is generated from MODIS NDVI datasets to evaluate vegetation health and identify areas affected by drought-related stress. Based on the computed VCI values, the study categorizes the region into different drought severity classes and develops a thematic drought severity map to support environmental assessment and informed decision-making.

3. Study Area

Study Area: Bundelkhand Region, India

The study has been carried out in selected districts of the Bundelkhand region, which extends across the states of Uttar Pradesh and Madhya Pradesh. This region is well known for recurring drought events, irregular rainfall patterns, and its high dependence on agriculture, making it highly susceptible to drought impacts. Owing to these characteristics, Bundelkhand serves as an appropriate study area for evaluating drought severity through satellite-based vegetation analysis.

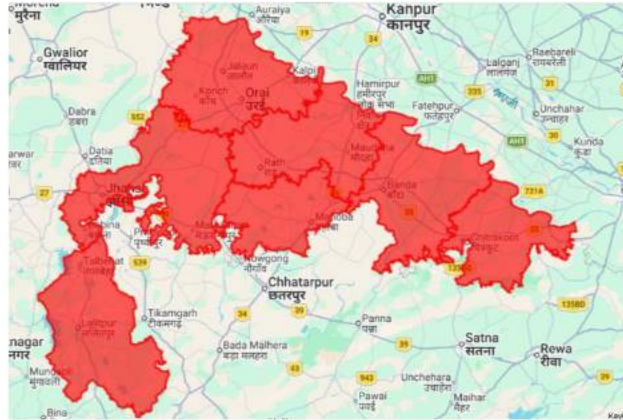


Figure 1. Area of Interest (AOI) representing the Bundelkhand Region.

4. Data Used

Satellite Data:

Parameter	Details
Satellite	MODIS
Product	MOD13Q1
Variable	NDVI
Temporal Coverage	2015–2025
Spatial Resolution	250 m
Platform	Google Earth Engine

Software

- Google Earth Engine (GEE)
- QGIS

Data Sources

1. MODIS MOD13Q1 NDVI Dataset

https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MOD13Q1

2. Google Earth Engine

<https://earthengine.google.com/>

3. FAO GAUL Administrative Boundaries:

[https://developers.google.com/earth-engine/datasets/catalog/FAO GAUL 2015 level2](https://developers.google.com/earth-engine/datasets/catalog/FAO_GAUL_2015_level2)

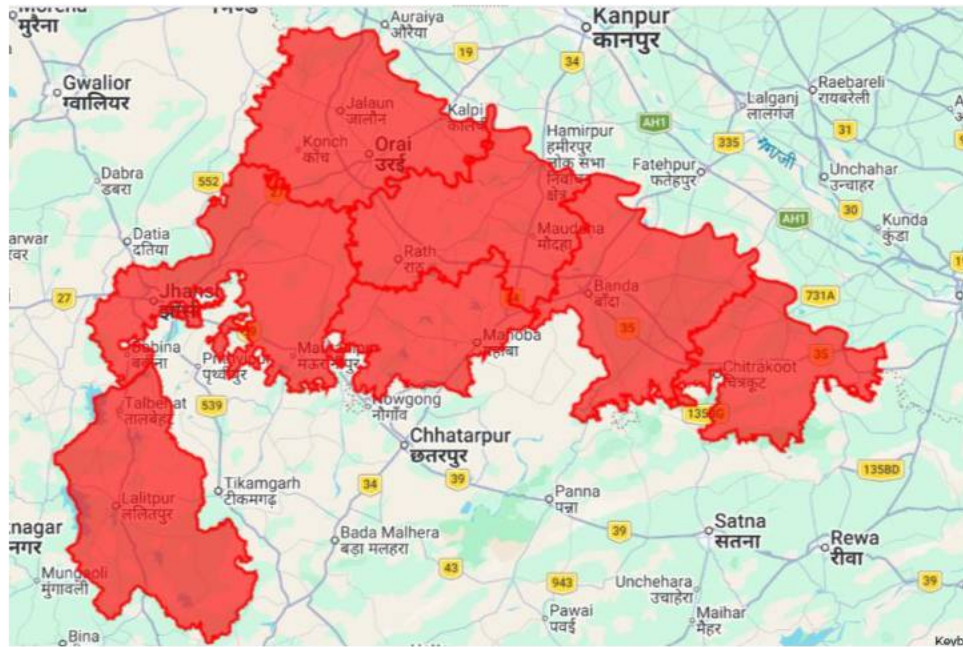
5. Methodology

Step 1: Area of Interest (AOI) Selection

The Bundelkhand region was identified as the Area of Interest (AOI) due to its frequent occurrence of drought and its vulnerability to water scarcity. Administrative boundary data for the selected districts were obtained from the FAO GAUL dataset and imported into Google Earth Engine (GEE). These boundaries were used to define the study area and perform the subsequent spatial analysis.

VCI Formula

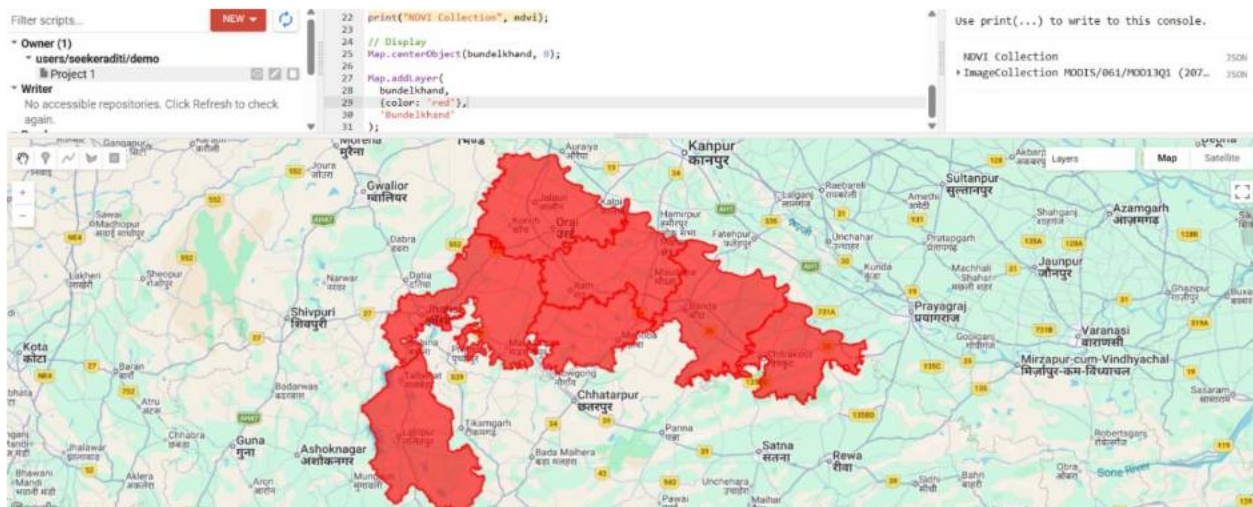
$$\text{VCI} = (\text{NDVI} - \text{NDVI}_{\min}) / (\text{NDVI}_{\max} - \text{NDVI}_{\min})$$



Step 2: Loading the MODIS NDVI Time-Series Dataset

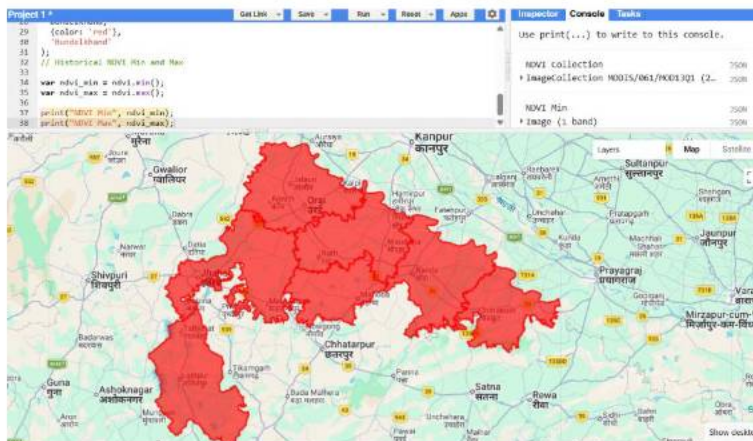
The MODIS Normalized Difference Vegetation Index (NDVI) time-series dataset was imported into Google Earth Engine (GEE) to examine temporal variations in vegetation conditions. Because MODIS provides satellite observations at regular intervals, it is well suited for long-term drought assessment and vegetation monitoring.

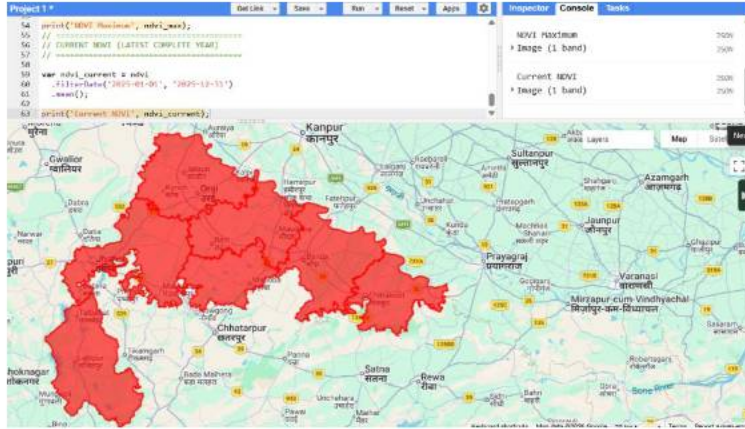
The image collection was filtered to match the selected Area of Interest (AOI) and the study period. These multi-temporal NDVI images served as the primary dataset for calculating the Vegetation Condition Index (VCI), which was subsequently used to evaluate vegetation stress and determine drought severity across the Bundelkhand region.



Step 3: Determination of NDVI Minimum and Maximum Values

To establish the range of vegetation conditions within the study area, the minimum (NDVImin) and maximum (NDVImax) values were derived from the multi-temporal MODIS NDVI image collection. For every pixel, the lowest and highest NDVI values recorded during the selected study period were identified. These values represent the historical limits of vegetation condition and provide the reference required for normalizing vegetation health. The generated NDVImin and NDVImax layers were then used in the computation of the Vegetation Condition Index (VCI), which enabled the assessment of vegetation stress and drought severity across the Bundelkhand region.





Step 4: Computation of the Vegetation Condition Index (VCI)

The Vegetation Condition Index (VCI) was calculated by combining the NDVI values from the MODIS time-series dataset with the corresponding NDVImin and NDVImax values for each pixel. This index standardizes the existing vegetation condition by comparing it with the historical minimum and maximum NDVI values, thereby reducing the influence of variations in vegetation type and geographical location. The computed VCI values range from 0 to 100, where lower values indicate higher vegetation stress and more severe drought conditions, while higher values reflect healthier vegetation and sufficient moisture availability. The resulting VCI layer served as the foundation for drought severity classification and thematic map preparation.

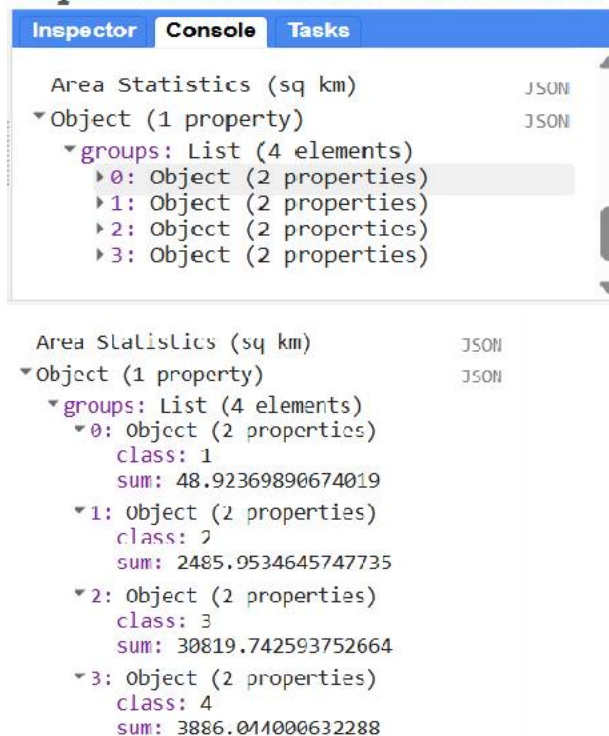
$$VCI = \frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}}$$

Step 5: Drought Severity Classification

The calculated VCI values were grouped into different drought severity categories based on predefined threshold ranges. Each class represents a specific level of vegetation condition and drought intensity, making it easier to interpret the spatial distribution of drought across the study area.

VCI Range	Category
< 0.2	Extreme Drought
0.2–0.4	Moderate Drought
0.4–0.6	Mild Drought
> 0.6	Healthy Vegetation

Step 6: Area Statistics Calculation



6. Results

VCI Map

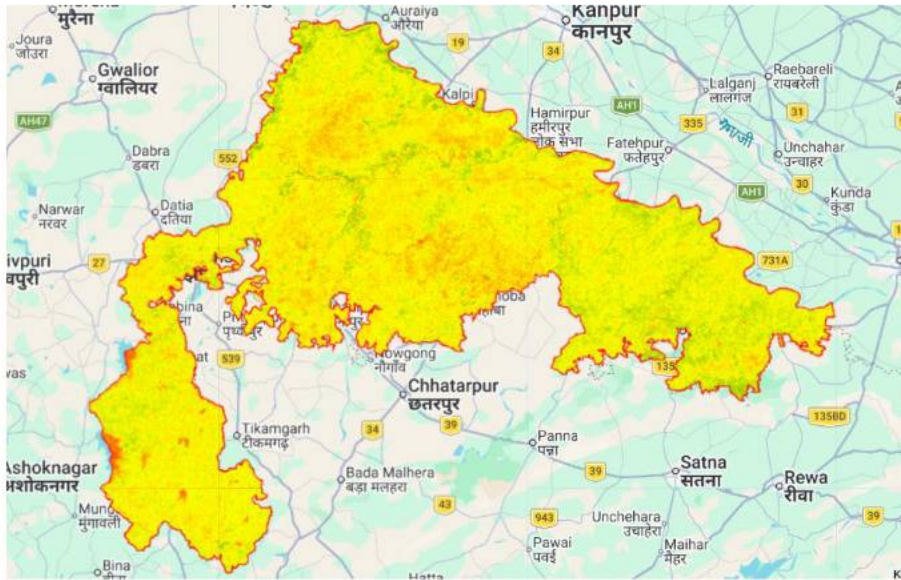


Figure 2. Vegetation Condition Index (VCI) Map

Drought Severity Map

Figure 3. Drought Severity Classification Map

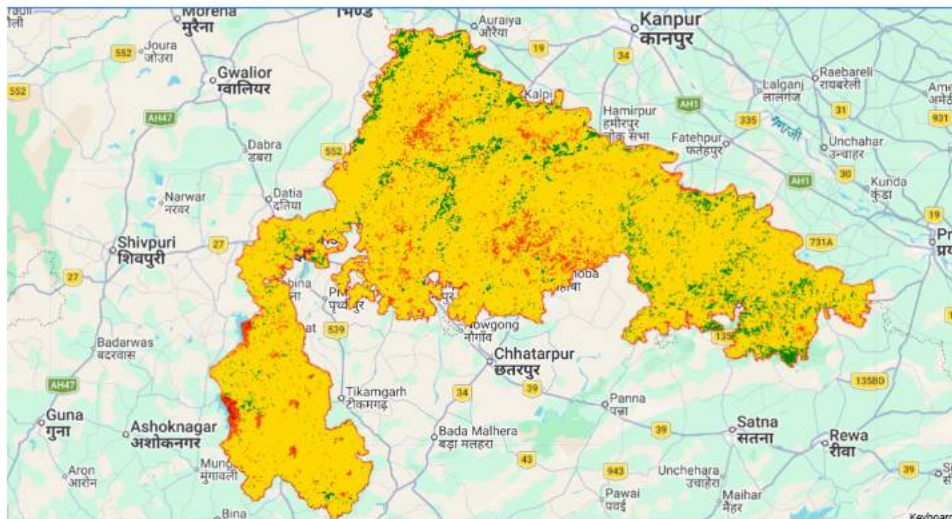


Figure 3. Drought Severity Classification Map

Area Statistics

Drought Class	Area (sq. km)
Extreme Drought	48.92
Moderate Drought	2485.95
Mild Drought	30819.74
Healthy Vegetation	3886.04

Interpretation

The drought severity assessment indicates that Mild Drought is the most widespread category within the Bundelkhand region, occupying approximately 30,819.74 sq. km. Moderate Drought covers about 2,485.95 sq. km, whereas the area affected by Extreme Drought is limited to 48.92 sq. km. In comparison, regions classified under Healthy Vegetation extend over nearly 3,886.04 sq. km.

These observations suggest that, although extreme drought is confined to a relatively small portion of the study area, a substantial part of the region continues to experience varying degrees of vegetation stress. This emphasizes the importance of regular drought monitoring and effective planning for sustainable resource management.

7. Conclusion

The outcomes of this study confirm that the Vegetation Condition Index (VCI) is an effective indicator for evaluating drought severity using satellite-derived vegetation information. By analyzing the MODIS NDVI time-series dataset, areas affected by vegetation stress and drought conditions across the Bundelkhand region were successfully identified.

The study demonstrates the significant role of remote sensing and GIS-based geospatial techniques in drought assessment, environmental monitoring, and supporting informed decision-making. To further improve the accuracy and reliability of future

drought assessments, additional variables such as rainfall and temperature datasets can be incorporated into the analysis.

8. References

- 1. MODIS MOD13Q1 NDVI Dataset.*
- 2. Google Earth Engine Platform.*
- 3. FAO GAUL Administrative Boundaries Dataset.*
- 4. Remote Sensing and GIS literature related to drought monitoring.*

END OF REPORT
